

PROBABILITY RULES

new 2.3–2.7 • old 4.1–4.6

ZONE A • THE FORMULAS

$$0 \leq P(A) \leq 1 \qquad P(S) = 1$$

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

mutually exclusive: $P(A \cap B) = 0$

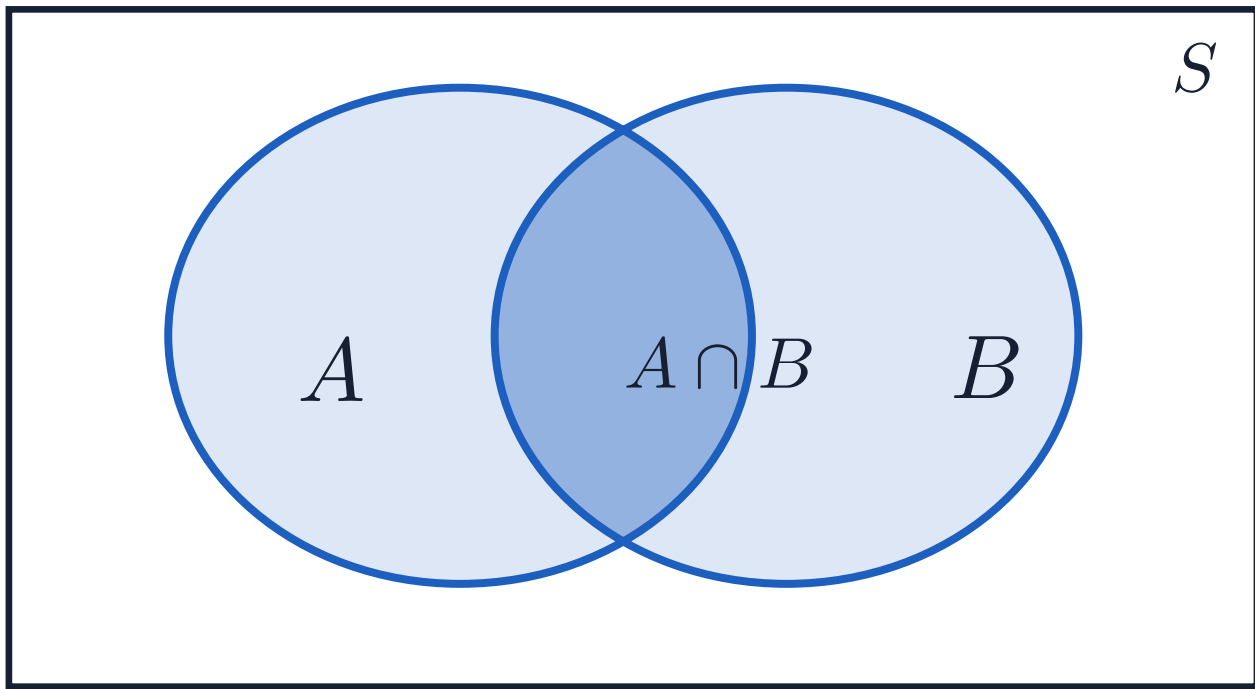
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \qquad P(B) > 0$$

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

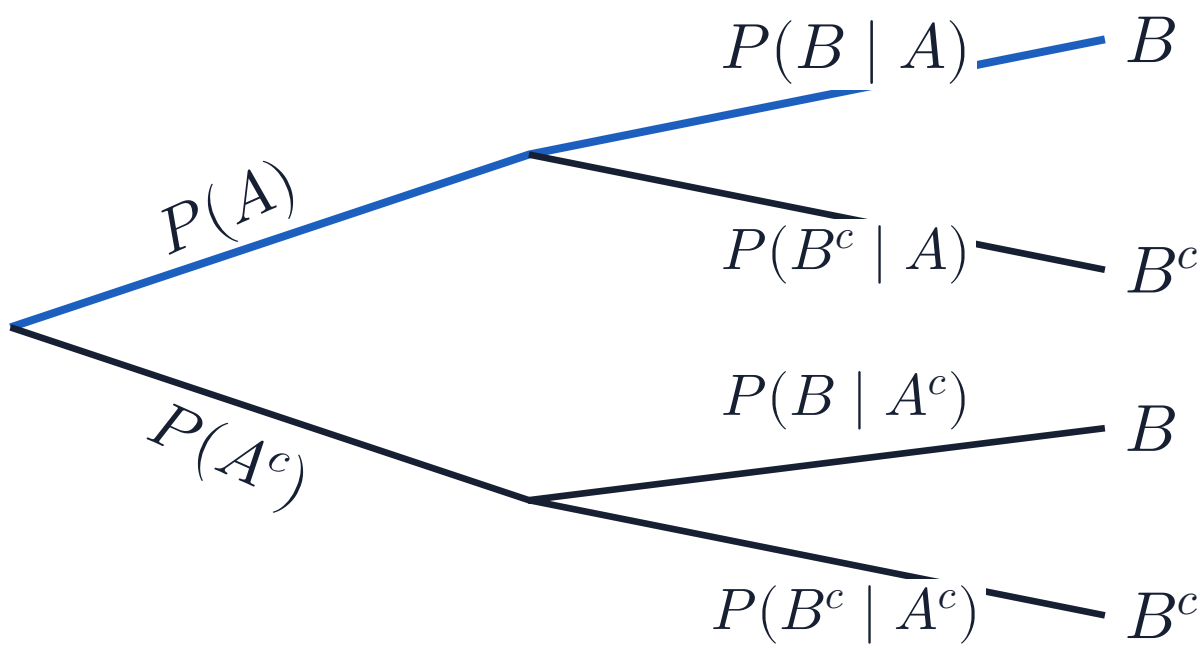
Independent $\iff P(A \mid B) = P(A)$
 $\iff P(A \cap B) = P(A) \cdot P(B)$

A conditional expression requires a positive probability for its given event.

OR: COUNT THE OVERLAP ONCE



AND: MULTIPLY ALONG A PATH



ZONE B • WHAT THE SYMBOLS MEAN

∪: **or** ∩: **and** |: **given**
 A^c : event A does **not** occur.
Outcome: one result.
Event: a set of outcomes.

Sample space S : all possible outcomes.
Law of large numbers: over many repetitions, an event's relative frequency approaches its probability.

ZONE C • THE SIMULATION RECIPE

1 • DESCRIBE ONE TRIAL

Use random digits or the calculator. Match the event and its chance.

2 • REPEAT

Run many trials. Count the trials meeting the success rule.

3 • ESTIMATE

$$P \approx \frac{\# \text{ successes}}{\# \text{ trials}}$$

State the event in context.

ZONE D • WATCH OUT

DISJOINT IS NOT INDEPENDENT

Mutually exclusive events are **never independent**, unless at least one has probability 0.

CHECK IT

Independence is checked, not assumed.